Technology Stack

Date	15 March 2026
Team ID	xxxxxx
Project Name	xxxxxx
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML5, CSS3, Bootstrap	Used to build a responsive, interactive, and user-friendly web interface. Bootstrap provides responsive layouts and modern UI components for better user experience.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python, Flask	Python is used for implementing the machine learning model, while Flask handles routing, user requests, prediction processing, and result rendering efficiently.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	CSV Dataset (HDI Dataset), Pickle (.pkl)	The HDI dataset is stored in CSV format for training and analysis. The trained Linear Regression model is saved as a Pickle file, allowing predictions without retraining.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Flask Local Server (Future: Render/Heroku)	The application is currently deployed on the Flask development server. It can be easily deployed on cloud platforms like Render or Heroku for public access.
5	**Version Control & CI/CD**	Git & GitHub	Git is used for source code version control, while GitHub is used for collaboration, project management, backup, and maintaining the project repository.

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, VS Code	Pandas and NumPy are used for data preprocessing, Matplotlib and Seaborn for data visualization, Scikit-learn for building the Linear Regression model, and VS Code for application development and debugging.